

BDC800 Proposal template

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Team Number: Group 2

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Title of the project: Forecasting Department-wise 911 Call Volume Using Time-Series Analysis

Summary of proposal:

Emergency response agencies rely on understanding call demand patterns to ensure operational readiness and effective staffing. This project proposes an analytical framework to explore historical 911 call data from Montgomery County, Pennsylvania, and to forecast short-term demand. The project focuses on predicting **hourly call volume for the next 24 hours**, separately for **EMS, Fire, and Traffic** departments.

The scope of the project is intentionally narrow and feasible. It includes exploratory data analysis (EDA) to identify temporal patterns and a time-series forecasting model to generate short-term predictions with uncertainty estimates. Optimization, dispatch decision-making, and real-time deployment are explicitly excluded. The outcome of the project is a validated forecasting pipeline that demonstrates how historical emergency call data can support proactive planning and preparedness decisions.

Team Information and responsibilities:

Responsibilities are divided using a pipeline-based structure to ensure clarity and accountability:

- **Dhruv Handa – Data Preparation & Aggregation**
Responsible for dataset cleaning, timestamp parsing, hourly aggregation, department categorization, handling missing hours, and preparing the final modeling dataset.
- **Aashish Sanjel – Exploratory Data Analysis & Visualization**
Responsible for analyzing temporal patterns (hourly, daily, weekly), generating visualizations, and interpreting department-wise trends.
- **Honey Jignesh Shah – Modeling & Evaluation**
Responsible for implementing the forecasting model, baseline comparison, time-based validation, and evaluating results using error metrics.
- **Raj Vipulbhai Jasani – Report Writing & Integration**
Responsible for integrating all sections, documenting methodology and results, and ensuring the final report aligns with academic and formatting requirements.

Overview of the problem:

911 call demand is inherently variable and influenced by time-based factors such as hour-of-day and day-of-week. While emergency call data is routinely collected, it is often used only for retrospective reporting rather than predictive analysis. Without quantitative forecasts, emergency

service planners may lack insight into short-term demand fluctuations, limiting their ability to prepare for predictable peaks in call volume.

The problem addressed in this project is the absence of a structured, data-driven approach for anticipating near-future 911 call demand at the departmental level. Specifically, the project seeks to transform raw historical call records into a form that allows for short-term forecasting, thereby enabling better-informed planning and preparedness discussions.

Specific Questions to Be Answered

This project is guided by the following specific questions:

1. How does 911 call volume vary by hour-of-day and day-of-week for EMS, Fire, and Traffic departments?
2. Can historical 911 call data be used to accurately forecast **hourly department-wise call volume 24 hours ahead**?
3. How does the performance of a time-series forecasting model compare to a strong seasonal baseline across different departments?

These questions ensure the project remains focused on descriptive insights and predictive performance rather than operational optimization.

Business Benefit of the Scientific or Technical Advances

The proposed project provides value by demonstrating how historical emergency call data can be leveraged for **decision support**. Short-term forecasts of department-wise call volume can assist emergency service managers in anticipating periods of higher demand and planning staffing levels accordingly.

From a broader perspective, the project illustrates how time-series analytics and uncertainty-aware forecasting can be applied to public-sector datasets. While the project does not claim real-time deployment, it highlights how predictive modeling can support proactive preparedness, reduce reactive decision-making, and inform resource planning discussions in emergency response environments.

Description of technical challenges:

Several technical challenges are inherent in this project:

- **Temporal aggregation:** Converting irregular, event-based call records into a continuous hourly time series while preserving meaningful patterns.
- **Department categorization:** Accurately classifying calls into EMS, Fire, and Traffic categories based on textual call descriptors.
- **Missing and zero-call periods:** Ensuring that hours with no recorded calls are correctly represented to maintain time-series continuity.

- **Seasonality modeling:** Capturing multiple overlapping seasonal patterns (hourly and weekly) in the data.
- **Evaluation rigor:** Designing a validation strategy that respects the temporal ordering of the data and avoids information leakage.

Addressing these challenges requires careful preprocessing, appropriate model selection, and transparent evaluation methods.

Proposed solution:

Dataset Description

The project uses the **MontcoAlert 911 Calls for Service** dataset, publicly available on Kaggle:

<https://www.kaggle.com/datasets/mchirico/montcoalert>

The dataset contains approximately 660,000 historical 911 call records from Montgomery County, Pennsylvania. Key variables include timestamps and call category information, which are sufficient for temporal aggregation and department-wise analysis.

Description of the Solution

The proposed solution follows a structured analytical pipeline:

1. Data Preparation:

- Parse timestamps and aggregate calls into hourly counts.
- Categorize calls into EMS, Fire, and Traffic.
- Represent data in a long-format structure (hour $\times$ department $\times$ call count).
- Explicitly include zero-call hours.

2. Exploratory Data Analysis:

- Examine hourly, daily, and weekly patterns in call demand.
- Compare temporal behavior across departments.

3. Forecasting Model:

- Implement a Prophet time-series forecasting model to predict hourly call volume 24 hours ahead for each department.
- Generate point forecasts and 80% prediction intervals to quantify uncertainty.

4. Baseline and Validation:

- Compare forecasts against a seasonal naïve baseline.
- Use a time-based holdout strategy with the last four weeks of data reserved for testing.
- Evaluate performance using Mean Absolute Error (MAE) as the primary metric and RMSE as a secondary metric.

Describe how the results will be delivered:

Results will be delivered in the form of:

- Visualizations showing historical patterns and forecast performance.
- Department-wise forecast plots comparing actual and predicted call volumes.
- Quantitative evaluation tables reporting MAE and RMSE for each department.
- A written analytical report interpreting results and discussing practical implications for emergency service planning.

Reference Dataset

McHirico. (n.d.). *MontcoAlert (Montgomery County, PA) 911 calls dataset* [Data set]. Kaggle. <https://www.kaggle.com/datasets/mchirico/montcoalert>